File Handling

```
import part of programming to create, read, update and delete files
handle data by open() built-in function
```

- "r" Read, for reading, if file doesn't exist return Error
- "a" Append, for appending, creates a file if doesn't exist
- "w" Write, for writing, creates a file if doesn't exist
- "x" Create, for creating a file, error if file exist
- "t" Text, Text Mode, for encoding to characters
- "b" Binary, Binary Mode, to treat data as bytes(images)

Reading

```
f = open("filename.txt")
txt = f.read(number_of_characters_to_read_optional)
#OR
txt = f.readline() # only the first line
#OR
txt = f.readlines() # return list of lines
#OR
txt = f.read.splitlines # return list of lines
```

Open a File with With Method

```
with open("filename.txt") as f:
    code
```

Writing and Updating

```
# a to append to end of the line
with open("filename.txt", "a") as f:
    f.write("text")

# w to overwrite existing content
with open("filename.txt", "w") as f:
    f.write("text")
```

Delete Files by OS

```
import os
os.remove("file.txt") # if file doesn't exist, raise an error

#OR
if os.path.exists("file.txt"):
    os.remove("file.txt")
else:
    code
```

File with txt Extension

File with json Extension

```
Javascript Object Notation, stringified JavaScript Object or Python
Dictionary
```

```
file_json = "{"..": ".."}"
```

| JSON to Dictionary

```
import json
file_json = "{}"
file_dct = json.loads(file_json)
```

| Dictionary to JSON

```
import json
file_dct = {}
file_json = json.dumps(file_dct, indent=4)
# indentation can be 2, 4, 8 for design purpose
```

| Saving JSON File

```
with open("file.json", "w", encoding="utf-8") as f:
    json.dump(data_to_write, ensure_ascii=False, indent=..)

# encoding is about how characters are stored in bytes
# ensure_ascii = False means to convert non ascii to escape sequences
```

File with CSV Extension

```
stands for comma separated value
store tabular data (spreadsheet or database)
```

```
common in data science
```

| CSV Style

```
"name", "country", "skills"
"Jevon", "Indonesia", "Python"
```

| Read CSV

```
with open..
    csv_reader = csv.reader(f, delimiter=", ")
    for row in csv_reader:
        row[0] = name
```

File with xlsx Extension

```
# first install xlrd package
import xlrd
excel_book = xlrd.open_workbook("file.xlsx")
print(excel_book.nsheets) # how many sheets exist
print(excel_book.sheet_names) # names of all worksheets
```

File with xml Extension

```
looks like html
tags are not predefined
```

```
<?xml version="1.0"?> #xml declaration
<root_name attribute1="">
#root to group all data and define document meaning
    <tag1></tag1>
    <tag2></tag2>
</root_name>
```

```
import xml.etree.ElementTree as ET
tree = ET.parse("file.xml")
root = tree.getroot()
print(root.tag)
print(root.attrib)
for child in root:
    print(child.tag)
```